

Precision Experimental Protocol for Energy-Budget-Resolved Testing of Nonequilibrium Electromagnetic Systems Application to Emergent Vacuum Response Theory (EVRT)

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Abstract

This work presents a precision experimental protocol for testing whether driven, nonequilibrium electromagnetic systems can produce measurable mechanical responses consistent with a fully constrained energy budget. Building on a thermodynamic energy-budget framework for Emergent Vacuum Response Theory (EVRT), the protocol defines a measurement architecture for quantifying input power, stored electromagnetic energy, mechanical output, environmental coupling, and loss channels in high-voltage asymmetric capacitor-coil systems. A torsion-balance-based apparatus is specified together with instrumentation requirements, calibration procedures, time-averaging methods, uncertainty propagation, and artifact-control strategies. The protocol emphasizes null-configuration validation, reproducibility, and residual interpretation within a pre-defined statistical criterion. The objective is not to demonstrate violations of conservation laws or free-energy generation, but to determine whether any residual mechanical or energetic signal remains after all identifiable contributions are quantified within uncertainty bounds. This provides a falsifiable pathway for evaluating EVRT-class hypotheses while preserving thermodynamic consistency and experimental restraint.

Keywords: EVRT; nonequilibrium electromagnetic systems; precision measurement; torsion balance; energy budget; residual force; artifact control; uncertainty analysis; high-voltage systems; experimental protocol

1 Introduction

Driven electromagnetic systems with asymmetric field geometry, dielectric boundaries, resonant structures, or high-voltage excitation can generate small mechanical signals through many ordinary mechanisms. These mechanisms include electrostatic attraction, corona discharge, ion wind, thermal drift, Lorentz-force coupling, radio-frequency leakage, vibration, acoustic excitation, and charge migration on nearby surfaces. For this reason, any claim of an anomalous mechanical response in such systems must be evaluated through a complete energy budget and a disciplined experimental protocol.

This paper provides a practical protocol for energy-budget-resolved testing of EVRT-class systems. It is intended as the experimental companion to a thermodynamic framework in which any residual term is treated as an uncertainty-bounded diagnostic quantity rather than evidence of new

physics by default [8]. The goal is to specify how each term in the device-level power balance can be measured, bounded, cross-checked, and interpreted.

The protocol is designed around a conservative principle: an observed mechanical deflection is meaningful only if it survives calibration, null configurations, artifact discrimination, and uncertainty propagation. The experimental objective is therefore not to demonstrate reactionless propulsion, free-energy generation, or violation of conservation laws. Instead, it is to determine whether a driven nonequilibrium electromagnetic structure exhibits a reproducible residual signal that cannot be attributed to known stored-energy changes, mechanical output, environmental coupling, or parasitic loss channels within the measurement uncertainty.

2 Measurement Philosophy

The experimental program is organized around the device-level power balance

$$P_{\text{in}} = \frac{dU_{\text{stored}}}{dt} + P_{\text{mech}} + P_{\text{loss}} + P_{\text{res}}, \quad (1)$$

where P_{in} is the measured input power, U_{stored} is stored electromagnetic and polarization energy, P_{mech} is mechanical output power, P_{loss} is the aggregate of identified loss channels, and P_{res} is the residual diagnostic term.

The null-consistent condition is

$$P_{\text{res}} = 0 \pm \sigma_{\text{res}}, \quad (2)$$

where σ_{res} is the combined standard uncertainty associated with the residual estimate. A provisional residual may be considered only when

$$|P_{\text{res}}| > n\sigma_{\text{res}}, \quad (3)$$

where n is selected before data collection. In exploratory testing, $n = 3$ may be used as a screening threshold; stronger claims require independent replication, blind controls, and systematic error studies.

Guiding Principle.

All observed mechanical effects must be traceable to an uncertainty-bounded energy budget before any physical interpretation is assigned. A nonzero residual is treated first as a metrological problem and only later, if reproducible under stringent controls, as a candidate physical anomaly.

3 Representative Apparatus

The representative apparatus consists of a lightweight asymmetric electromagnetic test article mounted on a sensitive torsion balance. The test article may include high-voltage electrodes, low-loss dielectric supports, optional magnet-wire or coil elements, and resonant or pulsed drive capability. The device is placed in a controlled environment with electrical shielding, thermal monitoring, and vibration isolation.

3.1 Core Components

The minimum experimental system should include:

- asymmetric high-voltage electrodes with smooth edges to reduce localized corona and uncontrolled field enhancement;

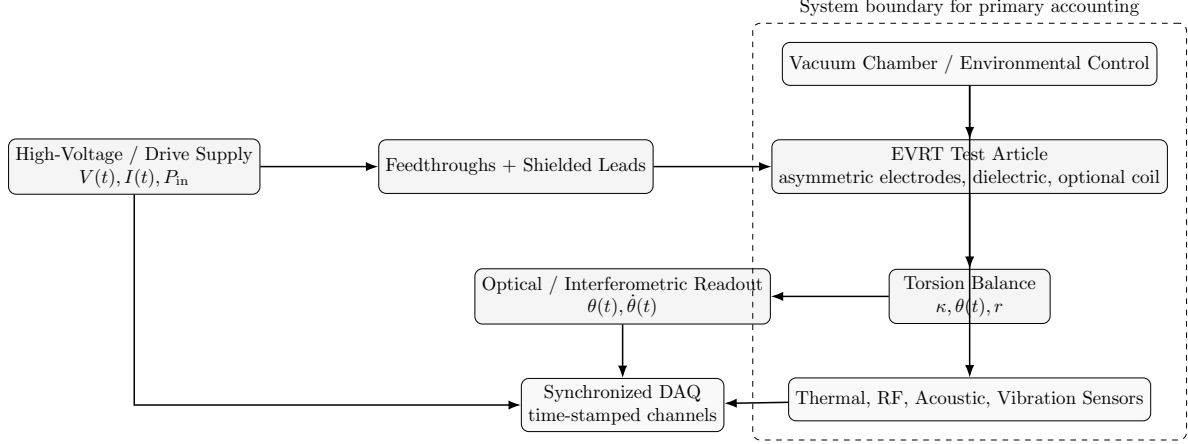


Figure 1: Representative apparatus for energy-budget-resolved EVRT-class testing. The test article is mounted on a torsion balance while electrical input, mechanical response, environmental variables, and artifact monitors are recorded with synchronized acquisition.

- a lightweight, low-loss dielectric support structure;
- optional coil or resonant element with independently measurable inductance and current;
- a torsion balance, microforce sensor, or equivalent force-measurement platform;
- optical, capacitive, or interferometric angular displacement readout;
- synchronized voltage, current, thermal, vibration, acoustic, and RF monitoring channels.

3.2 Drive Regime Classification

The behavior of EVRT-class systems depends strongly on the drive regime:

- **DC operation:** Dominated by steady-state fields and thermal effects.
- **Pulsed operation:** Transient charging, dielectric relaxation, and switching artifacts become significant.
- **Resonant operation:** Phase relationships between voltage and current must be carefully controlled; small phase errors can produce incorrect power estimates.

Each regime requires distinct calibration and artifact-control strategies.

3.3 Torsion Balance Configuration

For a torsion balance, the measured torque is

$$\tau = \kappa\theta, \quad (4)$$

where κ is the torsion constant and θ is the angular displacement. The equivalent force at arm length r is

$$F = \frac{\kappa\theta}{r}. \quad (5)$$

The mechanical power is

$$P_{\text{mech}} = \tau\omega = \kappa\theta\dot{\theta}, \quad (6)$$

while the quasi-static stored torsional energy is

$$E_{\text{mech}} = \frac{1}{2}\kappa\theta^2. \quad (7)$$

3.4 System Boundary

The system boundary must be defined before each run. At minimum, it includes the test article, its immediate electrical leads, field-coupled support structures, and any nearby objects capable of storing charge, carrying induced current, or experiencing electrostatic or magnetic coupling. Ambiguity in the system boundary can create false residuals by omitting stored energy, external coupling, or parasitic work pathways.

4 Measurement of Energy-Budget Terms

4.1 Input Power

Input power must be measured directly rather than inferred from nominal voltage alone. For high-voltage operation, the recommended measurement chain includes a calibrated high-voltage probe, a current monitor or shunt with appropriate bandwidth, and a synchronized data-acquisition system. The input power is evaluated as

$$P_{\text{in}}(t) = V(t)I(t), \quad (8)$$

with the time-averaged value

$$\bar{P}_{\text{in}} = \frac{1}{T} \int_{t_0}^{t_0+T} V(t)I(t) dt. \quad (9)$$

For pulsed or resonant operation, bandwidth limitations and probe phase response must be calibrated or bounded.

4.2 Stored Electromagnetic Energy

The stored-energy term is approximated by

$$U_{\text{stored}} = \frac{1}{2}C(t)V(t)^2 + \frac{1}{2}L(t)I(t)^2 + U_{\text{pol}}(t), \quad (10)$$

where $C(t)$ is effective capacitance, $L(t)$ is effective inductance, and $U_{\text{pol}}(t)$ is dielectric polarization energy. Measurements should include capacitance as a function of geometry and voltage, inductance as a function of frequency and configuration, and dielectric loss characterization when applicable.

The relevant power contribution is not the stored energy itself but its time derivative,

$$\dot{U}_{\text{stored}} = \frac{dU_{\text{stored}}}{dt}. \quad (11)$$

Transient charging and dielectric relaxation can dominate this term, especially during turn-on, turn-off, and polarity reversal.

4.3 Mechanical Output

Mechanical output is derived from calibrated angular displacement. The signal-processing pipeline should estimate $\theta(t)$, $\dot{\theta}(t)$, and the quasi-static deflection after subtracting baseline drift. The mechanical contribution should be reported both as force and as energy or power, depending on whether the measurement is dynamic or quasi-static.

4.4 Loss Channels

The aggregate loss term is

$$P_{\text{loss}} = P_{\text{Joule}} + P_{\text{dielectric}} + P_{\text{corona}} + P_{\text{RF}} + P_{\text{thermal}} + P_{\text{vib}} + P_{\text{leak}}. \quad (12)$$

Each contribution should be measured directly when possible or bounded conservatively when direct measurement is unavailable.

Table 1: Recommended measurement channels for the energy budget.

Quantity	Measurement method	Primary risk if omitted
P_{in}	HV probe, current monitor, true power calculation	Misestimated energy input
$\dot{U}_{\text{stored}}$	Capacitance/inductance characterization, transient tracking	Charging or dielectric relaxation mistaken for force
P_{mech}	Optical lever, interferometry, torsion calibration	Deflection not converted to energy scale
P_{thermal}	Thermistors, IR camera, dummy heater controls	Thermal drift or convection artifacts
P_{RF}	Spectrum analyzer, shielded load comparison	RF leakage or induced-current coupling
P_{corona}	Pressure sweep, current spikes, acoustic/optical monitoring	Ion wind or plasma forces
P_{vib}	Accelerometers, isolation platform, off-device dummy loads	Mechanical coupling mistaken for signal

Table 2: Target measurement resolution for key quantities in EVRT-class experiments.

Quantity	Target Resolution	Instrument
Voltage V	$< 1\%$	High-voltage probe
Current I	$< 1\%$	Current shunt / probe
Angle θ	$< 1 \mu\text{rad}$	Optical lever / interferometer
Force	$< 1 \text{ nN}$	Torsion balance
Temperature	$< 0.01 \text{ K}$	Thermistor / RTD
RF Power	mW-level	Spectrum analyzer

5 Calibration Procedures

5.1 Torsion Constant Calibration

The torsion constant κ should be calibrated using at least one independent method. Options include a known electrostatic torque, a calibrated microforce actuator, or a small known mass offset if the geometry permits. The calibration relation is

$$\tau_{\text{known}} = \kappa\theta, \quad (13)$$

with κ obtained from a linear fit over the range used in the experiment.

5.2 Force Calibration

A known electrostatic force can be applied in a controlled geometry to validate the mechanical readout. The calibration should be performed with the same optical path, sampling rate, damping conditions, and environmental configuration used during the primary experiment.

5.3 Electrical Calibration

Voltage probes, current monitors, and power meters must be calibrated across the relevant frequency range and voltage regime. For pulsed operation, phase shifts between voltage and current channels must be characterized because a small timing error can generate a false power estimate.

5.4 Environmental Baseline Calibration

Baseline drift should be measured with the device unpowered, with dummy loads, and with thermally matched controls. Thermal, vibration, and acoustic channels should be recorded during all baselines so that environmental correlations can be identified before powered operation.

6 Time Averaging and Data Acquisition

All power terms should be evaluated over a pre-defined measurement interval. The time-averaged value of a quantity $X(t)$ is

$$\overline{X} = \frac{1}{T} \int_{t_0}^{t_0+T} X(t) dt. \quad (14)$$

The averaging interval should be long enough to reduce high-frequency noise and switching artifacts, but short enough to avoid mixing distinct thermal or dielectric relaxation regimes. Each run should include synchronized timestamps for electrical, mechanical, thermal, RF, acoustic, and vibration channels.

Timing and Synchronization. All measurement channels must be time-synchronized to within sub-millisecond accuracy. In pulsed or resonant operation, phase offsets between voltage and current measurements can produce incorrect power estimates and artificial residuals. Synchronization should be verified through calibration signals and timing alignment tests prior to experimental runs.

6.1 Expected Signal-to-Noise Regime

Representative EVRT-class signals are expected to lie in the nanonewton range or below. For example, a force scale of order 10^{-9} N may be comparable to thermal drift, electrostatic coupling, or vibration-induced forces in uncontrolled environments.

Thermal expansion, convection, and dielectric relaxation can generate forces in the range of 10^{-9} – 10^{-8} N under typical laboratory conditions. Similarly, vibration and acoustic coupling can introduce forces of comparable magnitude.

Therefore, any candidate signal must be interpreted as a small perturbation embedded within competing noise sources. This necessitates strict environmental control, repeatability, and quantitative uncertainty analysis before any residual interpretation is considered.

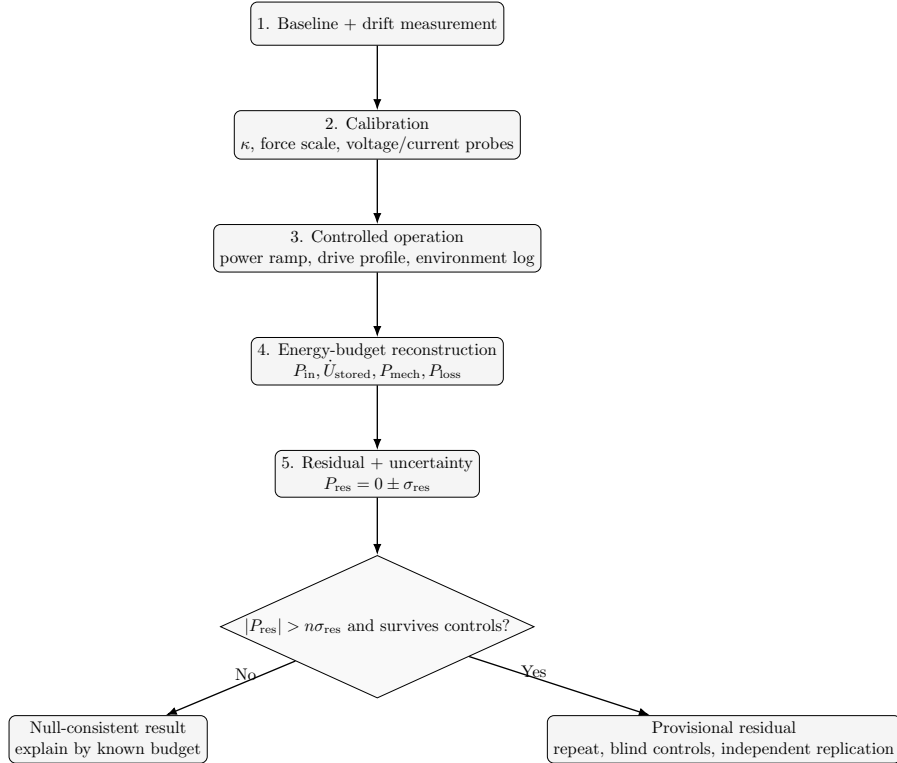


Figure 2: Measurement and interpretation workflow. A residual is considered only after calibration, controlled operation, energy-budget reconstruction, uncertainty propagation, and null-test validation.

7 Uncertainty Analysis

The residual is computed as

$$P_{\text{res}} = P_{\text{in}} - \left(\dot{U}_{\text{stored}} + P_{\text{mech}} + P_{\text{loss}} \right). \quad (15)$$

If the terms are treated as independent first-order contributions, then

$$\sigma_{\text{res}}^2 \approx \sigma_{P_{\text{in}}}^2 + \sigma_{\dot{U}}^2 + \sigma_{P_{\text{mech}}}^2 + \sigma_{P_{\text{loss}}}^2. \quad (16)$$

If correlations exist, covariance terms must be retained:

$$\sigma_{\text{res}}^2 = \sum_i \left(\frac{\partial P_{\text{res}}}{\partial x_i} \right)^2 \sigma_{x_i}^2 + 2 \sum_{i < j} \left(\frac{\partial P_{\text{res}}}{\partial x_i} \right) \left(\frac{\partial P_{\text{res}}}{\partial x_j} \right) \text{Cov}(x_i, x_j). \quad (17)$$

A residual should be reported as $P_{\text{res}} \pm \sigma_{\text{res}}$ together with the averaging interval, calibration state, null-test results, and environmental conditions. The uncertainty estimate should be validated through repeated measurements and, where possible, independent calibration cross-checks.

8 Null Tests and Control Configurations

Null configurations are required before any residual interpretation. The expected residual should be zero by construction for at least the following cases:

- **Power-off baseline:** quantifies drift, vibration, and thermal background.
- **Dummy load:** preserves electrical loading while removing the asymmetric field structure.
- **Symmetric geometry:** tests whether electrode asymmetry is necessary for the measured response.
- **Polarity reversal:** checks whether the signal changes sign, remains unchanged, or follows an artifact pattern.
- **Distance scaling:** tests electrostatic coupling to nearby surfaces and grounded structures.
- **Pressure sweep:** distinguishes vacuum-stable effects from ion wind, corona, and gas-mediated forces.

A candidate signal must survive these controls before being classified as a provisional residual.

9 Artifact Discrimination Strategy

Table 3 summarizes major artifact classes and corresponding rejection tests. These checks should be treated as required components of the protocol rather than optional diagnostics.

Scaling Behavior. A physically meaningful signal should exhibit consistent scaling with input power, field strength, or geometry. In contrast, many artifact sources exhibit non-physical or inconsistent scaling behavior. Scaling tests therefore provide an additional diagnostic tool for distinguishing genuine effects from environmental or instrumental artifacts.

10 Experimental Sequence

A recommended run sequence is:

1. define the system boundary and document geometry;
2. calibrate the torsion constant and readout sensitivity;
3. calibrate voltage, current, and timing channels;

Table 3: Artifact discrimination matrix for EVRT-class experiments.

Artifact	Mechanism	Discrimination test
Ion wind / EHD flow	Ionized gas and charge transport produce thrust-like forces	Vacuum operation, pressure sweep, polarity reversal
Corona discharge	Asymmetric plasma forces and acoustic emission	Dark testing, current-spike monitoring, optical/acoustic sensors
Thermal drift	Expansion, convection, radiometric effects	Dummy heaters, thermal imaging, time-lag analysis
Electrostatic coupling	Attraction to grounded or charged surroundings	Guard electrodes, distance scaling, shielding
RF leakage	Momentum transfer or induced currents in surroundings	Faraday shielding, spectrum monitoring, load substitution
Lorentz-force coupling	Current-carrying leads interact with fields	Twisted-pair routing, battery isolation, field mapping
Vibration / acoustic coupling	Mechanical excitation from supplies or discharge	Accelerometers, isolation stage, off-device dummy load
Dielectric relaxation	Slow charge migration and delayed response	Material comparison, discharge protocol, humidity control

4. record unpowered baseline drift;
5. record dummy-load and symmetric-geometry null tests;
6. apply controlled power ramps while recording all channels;
7. repeat operation under polarity reversal and pressure variation;
8. reconstruct the energy budget and compute P_{res} ;
9. propagate uncertainty and evaluate the pre-defined residual criterion;
10. repeat on independent days and, if possible, with independent apparatus configurations.

11 Interpretation Criteria

A measured response should be considered null-consistent if

$$P_{\text{res}} = 0 \pm \sigma_{\text{res}}. \quad (18)$$

A result should be considered a provisional residual only if all of the following are satisfied:

1. $|P_{\text{res}}| > n\sigma_{\text{res}}$ using a pre-defined threshold;
2. the signal is reproducible across repeated runs;
3. the signal survives null configurations;

4. the signal is stable under expected reversal or symmetry tests;
5. environmental channels do not correlate with the measured response;
6. independent calibration checks remain valid before and after the run.

Even then, the correct interpretation is not immediate evidence of new physics. The result should be reported as a candidate residual requiring further systematic testing.

12 Discussion

The protocol reframes EVRT-class testing as a precision measurement problem rather than a claim-driven demonstration. This framing is essential because high-voltage asymmetric systems can produce small forces through many conventional pathways. By requiring energy-budget reconstruction, uncertainty propagation, and null-test validation, the experiment becomes falsifiable in a way that is compatible with thermodynamics and classical electrodynamics.

The most important practical outcome is the separation of three categories: null-consistent behavior, conventional artifact behavior, and provisional residual behavior. Only the third category warrants deeper investigation, and only after environmental coupling, stored-energy transients, and systematic errors are aggressively constrained.

13 Conclusion

A precision experimental protocol has been developed for energy-budget-resolved testing of nonequilibrium electromagnetic systems relevant to EVRT. The protocol specifies a torsion-balance apparatus, measurement channels, calibration procedures, time-averaged data acquisition, uncertainty propagation, null tests, and artifact discrimination methods. The central diagnostic is the residual power term P_{res} , which must be evaluated relative to σ_{res} and interpreted conservatively.

The protocol is designed such that a null result provides meaningful constraints on EVRT-class hypotheses, while any residual result must satisfy increasingly stringent levels of experimental validation. In this way, both positive and null outcomes contribute to the systematic refinement of the experimental and theoretical framework.

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Data and Code Availability

No experimental data are reported in this manuscript. The protocol, equations, and figures are intended to guide future experimental design and interpretation. Supporting repository materials may be maintained at <https://github.com/HitojinKyoshi/evrt-research>.

Conflict of Interest Statement

The author declares no competing financial interests.

References

- [1] Herbert B. Callen. *Thermodynamics and an Introduction to Thermostatistics*. Wiley, 2nd edition, 1985.
- [2] John David Jackson. *Classical Electrodynamics*. Wiley, 3rd edition, 1998.
- [3] David J. Griffiths. *Introduction to Electrodynamics*. Cambridge University Press, 4th edition, 2017.
- [4] Edward M. Purcell and David J. Morin. *Electricity and Magnetism*. Cambridge University Press, 3rd edition, 2013.
- [5] F. Reif. *Fundamentals of Statistical and Thermal Physics*. McGraw-Hill, 1965.
- [6] Dilip Kondepudi and Ilya Prigogine. *Modern Thermodynamics: From Heat Engines to Dissipative Structures*. Wiley, 2nd edition, 2014.
- [7] Philip R. Bevington and D. Keith Robinson. *Data Reduction and Error Analysis for the Physical Sciences*. McGraw-Hill, 3rd edition, 2003.
- [8] Erick S. Sangalang. Thermodynamic Constraints and Energy Budget Framework for Nonequilibrium Electromagnetic Systems: Implications for Emergent Vacuum Response Theory. Zenodo, 2026. DOI: 10.5281/zenodo.20008382.
- [9] Erick S. Sangalang. Emergent Vacuum Response Theory (EVRT): Constraints, Numerical Simulations, and Experimental Pathways for Nonequilibrium Electrodynamics. Zenodo, 2026. DOI: 10.5281/zenodo.19891259.
- [10] Erick S. Sangalang. Systematic Artifact Analysis and Null Hypothesis Stress Testing for Emergent Vacuum Response Experiments. Zenodo, 2026. DOI: 10.5281/zenodo.19983226.
- [11] S. Schlamminger et al., “Recent measurements of the gravitational constant using torsion balances,” *Reviews of Modern Physics*, vol. 90, 2018.